

# A Set-Cover Counterexample to Uniform Duality for Outer $L_\mu^p(\ell^\infty)$

A candidate full negative answer to arXiv:2001.05903

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**Status.** Candidate counterexample, likely valid. For every  $1 < p < \infty$ , it disproves uniformity in the finite setting for both the reverse Köthe-duality estimate and the quasi-triangle inequality at  $r = \infty$ .

## Source question

Fraccaroli [1] studies outer  $L_\mu^p(\ell^r)$  spaces on a finite set  $X$  equipped with an outer measure  $\mu$  and a strictly positive weight  $\omega$ . After Theorem 1.1, the paper states:

For  $1 < p < \infty$ ,  $r = \infty$ , the question about uniformity remains open.

Here uniformity concerns both the reverse dual estimate

$$\|f\|_{L_\mu^p(\ell^\infty)} \leq C_p \sup_{\|g\|_{L_\mu^{p'}(\ell^1)}=1} \sum_{x \in X} \omega(x) |f(x)g(x)| \quad (D)$$

and the countable quasi-triangle estimate

$$\left\| \sum_j f_j \right\|_{L_\mu^p(\ell^\infty)} \leq C_p \sum_j \|f_j\|_{L_\mu^p(\ell^\infty)}, \quad (T)$$

with  $C_p$  independent of  $(X, \mu, \omega)$ . The opposite direction in (D) follows from the established outer Hölder and  $\ell^1$ -domination inequalities [1, Propositions 8.4 and 8.6]; see also Do-Thiele [2].

## Definitions

For  $r = \infty$ , the size is

$$\ell^\infty(f)(A) = \sup_{x \in A} |f(x)|,$$

and the superlevel outer measure is simply  $\mu(\{|f| > \lambda\})$ . Consequently

$$\|f\|_{L_\mu^p(\ell^\infty)}^p = \int_0^\infty p\lambda^{p-1} \mu(\{|f| > \lambda\}) d\lambda. \quad (1)$$

In particular, for  $a \geq 0$  and  $A \subseteq X$ ,

$$\|a\mathbf{1}_A\|_{L_\mu^p(\ell^\infty)} = a \mu(A)^{1/p}. \quad (2)$$

## Proof intuition

The outer measure will be a set-cover cost. We choose  $N$  unit-cost covering sets so that every point belongs to exactly  $k$  of them, but covering the whole ground set requires  $N - k + 1$  of them. Each covering-set indicator then has norm one. Their sum, however, is the constant  $k$  on a set of outer measure  $N - k + 1$ , so its norm has an extra factor  $(N - k + 1)^{1/p}$ . Taking  $k = N/2$  makes this factor diverge. The proposed dual expression cannot hide the obstruction: it is subadditive, and the already-known Hölder half bounds it uniformly on every individual indicator.

in the sense of the following inequality,

$$(1.7) \quad \|f\|_{L^p(\ell^r)} \leq C \sup_{\|g\|_{L^{p'}(\ell^{r'})}=1} \|fg\|_{L^1(X,\omega)},$$

where the constant  $C$  is independent of  $f \in L^p(\ell^r)$ , and  $L^1(X,\omega)$  stands for the classical  $L^1$  space on  $X$  with the measure associated with the weight  $\omega$ .

**Theorem 1.1.** *Let  $0 < p, r \leq \infty$ . There exists a constant  $C = C(p, r)$  such that, for every finite set  $X$ , outer measure  $\mu$ , and strictly positive weight  $\omega$ , the following properties hold true.*

(i) *For  $0 < p = r \leq \infty$ , for every  $f \in L^p(\ell^p)$ ,*

$$\frac{1}{C} \|f\|_{L^p(X,\omega)} \leq \|f\|_{L^p(\ell^p)} \leq C \|f\|_{L^p(X,\omega)}.$$

(ii) *For  $1 < p \leq \infty, 1 \leq r < \infty$  or  $p = r \in \{1, \infty\}$ , for every  $f \in L^p(\ell^r)$ ,*

$$\frac{1}{C} \sup_{\|g\|_{L^{p'}(\ell^{r'})}=1} \|fg\|_{L^1(X,\omega)} \leq \|f\|_{L^p(\ell^r)} \leq C \sup_{\|g\|_{L^{p'}(\ell^{r'})}=1} \|fg\|_{L^1(X,\omega)}.$$

(iii) *For  $1 < p \leq \infty, 1 \leq r < \infty$  or  $p = r \in \{1, \infty\}$ , for every  $\{f_n\}_{n \in \mathbb{N}} \subseteq L^p(\ell^r)$ ,*

$$\left\| \sum_{n \in \mathbb{N}} f_n \right\|_{L^p(\ell^r)} \leq C \sum_{n \in \mathbb{N}} \|f_n\|_{L^p(\ell^r)}.$$

Therefore, for  $1 < p \leq \infty, 1 \leq r < \infty$  or  $p = r \in \{1, \infty\}$ , the outer  $L^p(\ell^r)$  quasi-norm is equivalent to a norm, the outer  $L^p(\ell^r)$  space is isomorphic to a Banach space, and it is the Köthe dual space of the outer  $L^{p'}(\ell^{r'})$  space.

The main point of the theorem is the uniformity of the constant in  $(X, \mu, \omega)$ . In fact, for every fixed finite setting, both statements in (ii), (iii) are verified by a certain constant also for  $p = 1, 1 < r \leq \infty$  or  $1 < p < \infty, r = \infty$ , and hence the final considerations of the theorem hold true as well. However, for  $p = 1, 1 < r \leq \infty$ , the constant is not uniform in  $(X, \mu, \omega)$ , and we exhibit a counterexample in Lemma 3.4. For  $1 < p < \infty, r = \infty$ , the question about uniformity remains open. The uniformity of the constant suggests that if an infinite setting is suitably approximated by finite restrictions, the same results could possibly be obtained through a limiting process.

Figure 1: Theorem 1.1 and the open endpoint on source PDF page 3.

## Counterexample theorem

**Theorem 1.** *Fix  $1 < p < \infty$ . There is no constant depending only on  $p$  for which (T) holds in every finite outer measure setting. There is likewise no constant depending only on  $p$  for which the reverse dual estimate (D) holds in every such setting.*

More quantitatively, for each  $m \geq 1$  there is a finite setting and  $2m$  nonnegative functions  $f_1, \dots, f_{2m}$  such that

$$\frac{\left\| \sum_{i=1}^{2m} f_i \right\|_{L_\mu^p(\ell^\infty)}}{\sum_{i=1}^{2m} \|f_i\|_{L_\mu^p(\ell^\infty)}} = \frac{1}{2} (m+1)^{1/p}. \quad (3)$$

If  $D_p$  denotes the supremum on the right of (D), then for the same sum  $F_m$ ,

$$\frac{\|F_m\|_{L_\mu^p(\ell^\infty)}}{D_p(F_m)} \geq \frac{(m+1)^{1/p}}{2H_p}, \quad (4)$$

Here  $H_p < \infty$  is independent of the finite setting and satisfies  $D_p(h) \leq H_p \|h\|_{L_\mu^p(\ell^\infty)}$  by the already-known estimates.

*Proof.* We first give the construction with integers  $N > k \geq 1$ . Write  $[N] = \{1, \dots, N\}$  and let

$$X = \binom{[N]}{k} = \{S \subseteq [N] : |S| = k\}.$$

For each  $i \in [N]$ , define the incidence set

$$E_i = \{S \in X : i \in S\}.$$

Give the covering family  $\mathcal{D} = \{E_1, \dots, E_N\}$  the pre-measure  $\sigma(E_i) = 1$ , and let  $\mu$  be its induced covering outer measure:

$$\mu(A) = \min\{|I| : I \subseteq [N], A \subseteq \bigcup_{i \in I} E_i\}. \quad (5)$$

This is an outer measure of exactly the type allowed in the source paper. Every subset is covered by the full family. Moreover, every singleton  $\{S\}$  has outer measure one: it is nonempty and is covered by  $E_i$  for any  $i \in S$ . Thus  $\mu$  is finite and strictly positive on singletons. Take the strictly positive counting weight  $\omega(S) = 1$ .

Each  $E_i$  is nonempty and covers itself, hence

$$\mu(E_i) = 1. \quad (6)$$

We also have

$$\mu(X) = N - k + 1. \quad (7)$$

Indeed,  $\{E_i : i \in I\}$  covers  $X$  exactly when every  $k$ -subset of  $[N]$  meets  $I$ . This fails exactly when the complement  $[N] \setminus I$  contains a  $k$ -subset. Hence it is a cover exactly when  $|[N] \setminus I| \leq k - 1$ , or equivalently  $|I| \geq N - k + 1$ .

Set  $f_i = \mathbf{1}_{E_i}$ . By (2) and (6),

$$\|f_i\|_{L_\mu^p(\ell^\infty)} = 1. \quad (8)$$

Every point  $S \in X$  belongs to exactly  $k$  of the incidence sets, so

$$F := \sum_{i=1}^N f_i = k \mathbf{1}_X. \quad (9)$$

Equations (2), (7), and (9) give

$$\|F\|_{L_\mu^p(\ell^\infty)} = k(N - k + 1)^{1/p}, \quad \sum_{i=1}^N \|f_i\|_{L_\mu^p(\ell^\infty)} = N. \quad (10)$$

Now set  $N = 2m$  and  $k = m$ . Dividing the two quantities in (10) gives (3), which tends to infinity with  $m$ . This proves the failure of a uniform constant in (T).

For completeness, define

$$D_p(h) = \sup_{\|g\|_{L_\mu^{p'}(\ell^1)}=1} \sum_{S \in X} \omega(S) |h(S)g(S)|.$$

This functional is subadditive. The uniform outer Hölder direction proved in the source gives

$$D_p(h) \leq H_p \|h\|_{L_\mu^p(\ell^\infty)} \quad (11)$$

for a constant  $H_p$  independent of the finite setting. Applying (11) to (8) and using subadditivity yields

$$D_p(F) \leq \sum_{i=1}^N D_p(f_i) \leq H_p N.$$

Together with (10), and again  $N = 2m$ , this is (4). The ratio diverges, so no setting-independent reverse constant in (D) exists.  $\square$

## Why this resolves the endpoint

The source asks about uniformity for the entire missing range  $1 < p < \infty$ ,  $r = \infty$ . Theorem 1 treats every exponent in that range, under the source's full finite-setting hypotheses, and disproves both requested uniform statements. No unproved condition is left. The example does not concern the positive upper-half-space theorem, whose additional geometric structure is absent here.

The construction also gives a cardinality lower bound. Since  $|X| = \binom{2m}{m} \leq 2^{2m}$ , the necessary quasi-triangle constants are at least a fixed  $p$ -dependent multiple of  $(\log |X|)^{1/p}$ . No optimality is claimed.

## Verification and novelty check

The proof uses only the exact set-cover computation in (7), the incidence identity in (9), the source definition (1), and the already-established Hölder half (11). The accompanying script exhaustively enumerates the covering subfamilies for  $2 \leq N \leq 10$  and checks (6)–(10). These finite checks are corroboration, not proof.

A bounded search on August 9, 2026 included the run’s four lightweight indexes; the exact arXiv id, title, and open-question phrase; combinations of “outer  $L^p$ ”, “ $\ell^\infty$ ”, “Choquet”, “finite outer measure”, “uniformity”, and “triangle inequality”; the author’s later outer- $L^p$  papers and current publication list; and the author’s 2022 thesis. The thesis repeats that this exact endpoint remains open. No later solution or this set-cover example was found. Novelty confidence is therefore moderate to high, subject to specialist review.

## Human-review recommendation

The direct quasi-triangle obstruction is elementary and should be checked first. A reviewer should then confirm that the normalization and direction of the source’s outer Hölder estimate give (11). If both checks pass, this is a full negative resolution of the stated uniformity question.

## References

- [1] M. Fraccaroli, *Duality for outer  $L_\mu^p(\ell^r)$  spaces and relation to tent spaces*, J. Fourier Anal. Appl. **27** (2021), Paper No. 67; arXiv:2001.05903, <https://arxiv.org/abs/2001.05903>.
- [2] Y. Do and C. Thiele,  *$L^p$  theory for outer measures and two themes of Lennart Carleson united*, Bull. Amer. Math. Soc. (N.S.) **52** (2015), 249–296.